

**Keywords:** 4D reconstruction • video understanding • feed-forward transformer • dynamic scene analysis

# Efficiently Reconstructing Dynamic Scenes One D4RT at a Time

*A Single Feedforward Transformer Jointly Infers Depth, Correspondences, and Camera Parameters — 18–300× Faster Than Prior State of the Art*

By Chuhan Zhang, Guillaume Le Moing, Skanda Koppula *et al.* (Google DeepMind, UCL, University of Oxford)

arXiv:2512.08924 • CVPR 2026 Best Paper Award

The traditional 4D reconstruction pipeline — depth estimator, optical flow network, SLAM system, fusion module — is replaced by a single feedforward transformer that jointly infers depth, spatio-temporal correspondences, and camera parameters from a monocular video. D4RT achieves 18–300× speedups over specialized baselines while matching or surpassing their accuracy, and was awarded Best Paper at CVPR 2026 from over 16,000 submissions.

## AT A GLANCE

**Problem:** 4D dynamic scene reconstruction requires chaining multiple specialized models, compounding errors and consuming excessive compute.

**Why it matters:** Autonomous vehicles, AR/VR, and robotic manipulation all need fast, accurate 4D understanding of moving scenes.

**Method:** A unified transformer with 4D positional encodings and query-based decoding that processes depth, correspondence, and camera estimation in a single forward pass.

**Result:** 18–300× speedup over per-task specialists; new state-of-the-art on depth, optical flow, and camera estimation.

**Evidence:** Evaluated on Dynamic Replica, Sintel, DAVIS, Spring, Kubric, and in-the-wild internet videos.

## The Big Picture

Recovering the 3D structure and motion of a dynamic scene from video is one of computer vision’s fundamental challenges. The canonical approach chains specialist models: run a monocular depth estimator frame by frame, compute optical flow between adjacent frames with a dedicated flow network, recover camera motion with a SLAM or structure-from-motion system, and then fuse. Each component is trained independently, optimized for its own metric, and blind to the outputs of the others. Errors

accumulate.

D4RT asks: what if a single feedforward network handles all three tasks jointly, letting depth inform correspondence and correspondence constrain camera estimates in a shared representational space? The answer is that it works, and that joint modelling substantially improves each sub-task while eliminating the latency of pipeline execution.

## Why Existing Approaches Fall Short

**Error compounding:** In a pipeline, depth estimation errors propagate into optical flow, which propagates into SLAM. Joint inference breaks this error chain by sharing evidence across tasks.

**Computational redundancy:** Each specialist extracts its own visual features, producing redundant computation. Sharing a backbone across tasks eliminates this.

**Inference latency:** Sequential pipeline execution is bounded below by the sum of individual model runtimes. A single feedforward model is bounded only by its own latency, which with modern transformers can be significantly smaller.

**Dense decoding cost:** Prior feedforward 4D methods decode predictions for all pixels in all frames simultaneously, making inference cost quadratic in video length. D4RT’s query-based decoding is linear in the number of queries requested.

## Core Mechanism

**Unified token sequence.** D4RT embeds all frames into a joint sequence of patch tokens, each annotated with a 4D positional encoding  $(x, y, t, \tau)$  where  $x, y$  are pixel coordinates,  $t$  is the frame index, and  $\tau$  denotes the task modality (depth, correspondence, or camera). A standard transformer stack then lets all tokens attend to each other via full self-attention, sharing information across frames and tasks.

**Query-based inference.** At test time, the user specifies a set of queries  $Q = \{(x_i, y_i, t_i, \tau_i)\}$ . Each query cross-attends to the full token sequence, producing a task-specific prediction at that pixel and frame. Inference cost is  $O(|Q| \cdot N)$  where  $N$  is the total token count, not  $O(N^2)$ .

**Training.** All three task losses are applied simultaneously in every forward pass on video data with ground-truth depth, flow, and camera labels.

## Technical Formulation

Let  $V \in \mathbb{R}^{T \times H \times W \times 3}$  be an input video. Patch embeddings produce context tokens:

$$\mathbf{Z} = \text{PatchEmbed}(V) + \text{PE}_{4D}(x, y, t, \tau),$$

where  $\text{PE}_{4D}$  is a learned positional encoding across all four dimensions simultaneously.

After transformer processing  $\mathbf{Z}' = \text{Transformer}(\mathbf{Z})$ , each query  $q_i = (x_i, y_i, t_i, \tau_i)$  is decoded via:

$$\hat{y}_i = \text{MLP}(\text{CrossAttn}(q_i, \mathbf{Z}')).$$

The training objective is:

$$\mathcal{L} = \lambda_d \mathcal{L}_{\text{depth}} + \lambda_c \mathcal{L}_{\text{corr}} + \lambda_{\text{cam}} \mathcal{L}_{\text{camera}},$$

where each loss is a task-appropriate combination of  $L_1$  regression and angular/rotation losses, evaluated over the queried subset.

### Learning or Inference Procedure

Training proceeds end-to-end on mixed video datasets with ground-truth annotations:

1. Sample a clip of  $T$  frames; embed all patches with 4D PE.
2. Sample random query pixels per task modality.
3. Compute joint loss  $\mathcal{L}$  across all modalities.
4. Update via AdamW with warmup and cosine decay.

At inference:

1. Embed the input video.
2. Specify queries for desired outputs (dense depth, sparse correspondences, camera poses).
3. Decode predictions in a single forward pass.

### What the Guarantee Says

D4RT does not provide formal convergence guarantees. The empirical claim is: a single feedforward model can match or surpass specialized baselines on each individual 4D sub-task while achieving 18–300 $\times$  wall-clock speedup, with the speedup increasing with video length due to query-based decoding’s linear scaling.

### Experimental Findings

Evaluated across seven benchmarks:

- **Dynamic Replica:** new state-of-the-art on depth and correspondence metrics.
- **Sintel:** competitive with RAFT on optical flow (EPE within 3%) at 100 $\times$  faster runtime.
- **DAVIS:** strong video object correspondence without domain-specific fine-tuning.
- **Spring & Kubric:** generalizes to synthetic benchmarks outside training distribution.
- **In-the-wild videos:** qualitatively coherent 4D reconstructions of complex internet video.

Speedup range: 18 $\times$  (short clips, dense queries) to 300 $\times$  (long videos, sparse queries).

### Ablations and Interpretation

**Joint vs. separate training:** Joint multi-task training improves each sub-task by 5–12% over single-task training, confirming beneficial task interaction.

**4D positional encoding:** Removing the frame-index dimension of PE degrades correspondence quality by 18%, confirming that temporal grounding is critical.

**Query-based vs. dense decoding:** Dense decoding provides marginally higher accuracy (<2%) at 10 $\times$  higher compute. Query-based decoding is preferable for deployment.

### Reference

Chuhan Zhang, Guillaume Le Moing, Skanda Koppula, *et al.* **Efficiently Reconstructing Dynamic Scenes One D4RT at a Time.** arXiv:2512.08924, December 2025. <https://arxiv.org/abs/2512.08924>